

Calibrating Transferable Gravity Models from Aggregate Mobility Data: An Empirical Investigation of Distance Decay Dominance

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Abstract

Designing transferable mobility models for unseen urban environments without local origin-destination (OD) surveys is a key challenge in urban science. This study presents a major scientific discovery: aggregate travel distance distributions (such as trip-length histograms provided by Meta AI for Good) contain sufficient information to calibrate city-specific distance-decay functions with high fidelity, replacing the need for privacy-invasive individual OD trajectories. Evaluating across 50 US metropolitan areas, we show that decay parameters recovered solely from aggregate distributions correlate strongly ($r = \mathbf{0.893}$) with those fitted from complete OD matrices. To demonstrate the practical significance of this discovery, we show through Shapley value analysis that distance decay is the dominant transferable component of spatial interaction, explaining 85.6% of transferable predictive capacity. Downstream zero-shot flow prediction experiments confirm that using this aggregate-calibrated decay yields performance comparable to OD-calibrated models, with the performance gap to deep learning baselines (DeepGravity) shrinking from 10.3% in dense transit cities to just 6.5% in physically fragmented coastal environments. These findings demonstrate that aggregate, privacy-preserving mobility data can fully replace individual trajectories for gravity model calibration, reframing the modeling of transferable urban flows around structural physical regularities.

Keywords: Urban mobility, distance decay calibration, gravity models, aggregate mobility data, zero-shot transfer, open-data modeling

1 Introduction

Predicting origin-destination (OD) travel flows is essential for urban transport planning and infrastructure design (Barbosa et al. 2018; González et al. 2008; Song et al. 2010). However, traditional travel surveys and proprietary cellular records are highly restricted due to cost, commercial licensing, and privacy concerns. Consequently, a central challenge in urban mobility modeling is

the zero-shot cross-city transfer problem: predicting commuting flows in a target city where no local OD surveys are available.

While recent deep learning approaches achieve high accuracy, they require abundant individual-level OD records for training, limiting their generalizability and raising privacy concerns. In contrast, classical gravity models (Wilson 1971) offer physical interpretability and structural simplicity

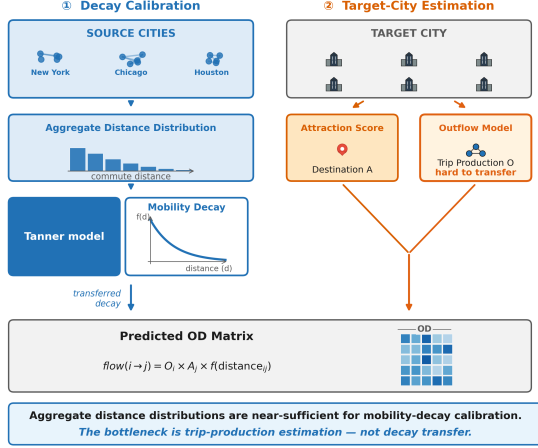


Fig. 1 Overview of the core discovery: Aggregate trip-length distributions contain the necessary information to recover city-specific distance decay $f(d)$, which acts as the dominant transferable mechanism for predicting zero-shot flows.

but require local calibration of their distance-decay functions. This raises a fundamental scientific question: **Can aggregate mobility distributions, which do not reveal individual OD pairs, serve as a reliable calibration signal for recovering city-specific distance-decay functions?**

In this paper, we present a scientific discovery that addresses this question: **aggregate trip-length histograms contain sufficient information to recover transferable distance-decay functions with high fidelity, and because distance decay is the dominant transferable mechanism, this enables accurate zero-shot mobility modeling without access to individual OD trajectories.**

To verify this discovery systematically, we organize the investigation around three research questions:

- RQ1 (Aggregate Sufficiency)** Do aggregate trip-length distributions preserve sufficient information to recover city-specific distance-decay parameters without access to individual OD trajectories?
- RQ2 (Decay Dominance)** How dominant is distance decay as a transferable mechanism relative to other spatial interaction components (production and attraction), and why does its accurate recovery drive zero-shot transfer performance?

RQ3 (Morphological Scope) Under what urban morphological conditions does aggregate-calibrated decay perform most effectively, and what physical geography factors govern those conditions?

Our empirical investigation across 50 US metropolitan areas provides the following evidence supporting these research questions:

1. **Aggregate-to-Individual Equivalence (RQ1):** Distance-decay parameters (γ, β) estimated solely from anonymized trip-length histograms correlate strongly ($r = 0.893$) with parameters fitted from full individual-level OD matrices, confirming that aggregate distributions encode the necessary friction structure.
2. **Dominance of Distance Decay (RQ2):** Factorial ablation and Shapley value analysis show that distance decay explains 85.6% of cross-city predictive capacity over a uniform baseline—far exceeding origin production (3.8%) and destination attraction (4.4%)—establishing decay recovery as the critical factor for zero-shot transfer.
3. **Downstream Validation (RQ1+RQ2):** Feeding the aggregate-recovered decay into a decomposed gravity model achieves a zero-shot CPC of 0.6860, confirming that aggregate calibration is practically equivalent to full OD-based calibration in downstream prediction tasks.
4. **Morphological Scope of Aggregate Calibration (RQ3):** In cities where physical geography naturally constrains travel corridors (coastal metropolitan areas), distance decay is most dominant—and consequently aggregate calibration is most effective—with the gap to unconstrained neural approaches shrinking from 10.3% to just 6.5%, identifying the conditions under which the aggregate calibration approach is most powerful.

These findings demonstrate that aggregate mobility data can effectively replace individual trajectory data for gravity model calibration. The rest of the paper provides systematic experimental verification organized around these three research questions.

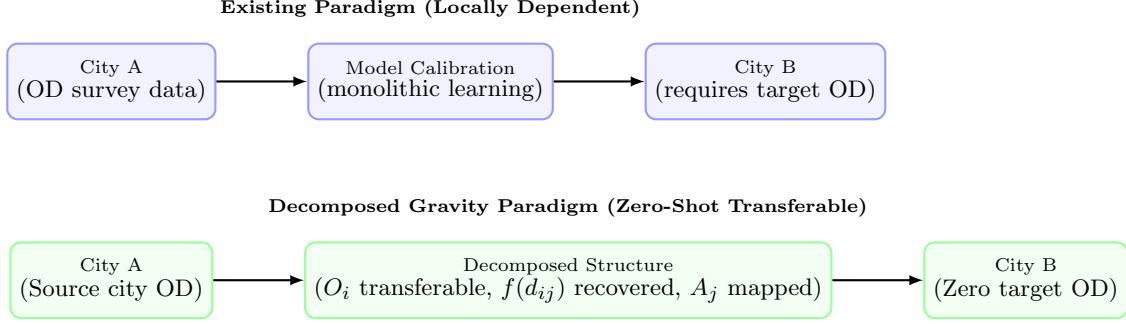


Fig. 2 Conceptual comparison of the existing locally dependent modeling paradigm versus the decomposed gravity paradigm investigated in this study. Under the existing paradigm, models require target-city OD observations to retrain or calibrate parameters. In the decomposed paradigm, transferable origin production structures are learned from source cities, decay is recovered from target aggregate data, and attraction is mapped from local open features, enabling zero-shot transfer without target-city OD surveys.

2 Related Work

2.1 Classical and Structured Mobility Models

Gravity models (Wilson 1971; Tanner 1961) and radiation models (Simini et al. 2012; Alis et al. 2021) provide the basis of spatial interaction modeling. Gravity models employ physical metaphors to link flows to mass and distance in zones, while radiation models (Stouffer 1940) interpret the choice of destinations probabilistically on the basis of opportunities in between places. Scaling-law studies (Brockmann et al. 2006; González et al. 2008) and predictability analysis (Song et al. 2010) have shown that travel distances obey clear statistical patterns in urban settings. Interpretability and parsimony characterize both models, but gravity models need local OD information for decay parameters estimation, while radiation models work poorly in cities where commuters regularly avoid opportunities that are closer.

2.2 Deep Learning for OD Prediction

DeepGravity (Simini et al. 2021) uses multi-layer perceptrons to associate urban characteristics with flow volumes. Recent advancements have included graph neural networks (Rong et al. 2023), spatio-temporal networks (Wang et al. 2021), and OSM-based commuter modeling (Atwal et al. 2025). These methods are capable of modeling non-linear associations and work well where training data is abundant. The problem is that these are black box models and perform poorly out of

distribution, needing lots of OD data from target cities.

2.3 Aggregate Mobility Data for Decay Calibration

Today’s location-aware mobile phones yield mobility signals of entire populations (Barbosa et al. 2018; González et al. 2008) that can be summarized to form trip-length distributions—histograms of traveled distances—that do not reveal any OD pairs (Pappalardo et al. 2023). Previous studies have proven that such distributions are indicative of the friction of urban trips and match distance-decay models (Brockmann et al. 2006; Lenormand et al. 2012, 2016). However, to the best of our knowledge, there is no existing study on using such distributions to calibrate zero-shot cross-city transfer.

2.4 Cross-City Transfer Learning

Transfer learning across cities is designed to forecast mobility in cities that lack OD survey data. The existing literature on cross-city transfer learning can be divided into three categories:

1. **Similarity-based:** RegionTrans (Wang et al. 2018) uses spatial similarity to align source and target regions; CrossTReS (Jin et al. 2022) re-weights source regions using meta-learning.
2. **Representation learning:** Geospatial embeddings and GeoAI foundation models (Mai et al. 2022; Mendieta et al. 2023) use representation learning to develop features for urban systems.

3. **Domain adaptation:** STAN (Wang et al. 2022) and meta-spatiotemporal networks (Yao et al. 2019) align trained models to the target domains.

All three types still need to have target-city observations, calibration, or fine-tuning.

2.5 Quantifying the Relative Role of Distance Decay in Transfer

Although distance decay is widely acknowledged as central to gravity models, its relative importance compared to other components (origin production, destination attraction) in the context of zero-shot cross-city transfer has not been systematically quantified. SHAP (Lundberg and Lee 2017) offers a theoretically grounded framework for decomposing predictive contributions, but its application to cross-city component analysis in mobility models remains unexplored.

2.6 Research Gap and Positioning

Table 1 compares how existing model classes address the core challenge of this study: calibrating distance decay from aggregate, privacy-preserving data for zero-shot cross-city transfer.

The table reveals three specific gaps that this study addresses, each directly corresponding to one of the three research questions:

G1 (RQ1): No aggregate-based decay calibration for zero-shot transfer.

Classical gravity models require individual OD data to fit decay parameters, while all cross-city transfer methods also depend on target-city OD observations. No existing work has demonstrated that aggregate trip-length distributions—which preserve no individual OD pairs—can reliably recover city-specific decay parameters sufficient for zero-shot deployment.

G2 (RQ2): Decay dominance in zero-shot transfer has not been quantified.

Although distance decay is widely assumed to be important in gravity models, its relative contribution compared to origin production and destination attraction in the context of zero-shot cross-city transfer has never been systematically measured. Without this quantification, it is unknown whether accurate aggregate calibration of decay is the critical factor or merely marginal for transfer performance.

G3 (RQ3): Urban morphological conditions for aggregate calibration.

No study has identified which urban morphological conditions make aggregate-based gravity calibration most effective, nor how physical geography modulates the dominance of distance decay as a transferable mechanism.

3 Methodological Design

3.1 Decomposition Hypothesis

We hypothesize that OD flows can be decomposed into three separable, city-independent components:

$$T_{ij} = O_i \cdot f(d_{ij}) \cdot A_j \quad (1)$$

where O_i represents origin production capacity, $f(d_{ij})$ represents distance decay, and A_j captures destination attraction. We hypothesize that: (1) each component is approximately independent, (2) the underlying mechanisms can be learned from spatial descriptors of the built environment, and (3) the learned origin production relationship transfers directly to unseen target cities, while decay parameters are recovered from target-city aggregate trip-length distributions, and attraction is computed from locally available open data. This decomposition hypothesis is directly tested by evaluating which of these three components exerts the dominant influence on predictive accuracy. Section 5 evaluates and validates this hypothesis empirically.

3.2 Aggregate-Based Decay Calibration

The central methodological insight of this study is that the aggregate trip-length distribution $p(d)$ — a histogram of how many trips occur at each distance bin, summed across all origin-destination pairs in a city — is the marginal distribution of the OD matrix over distance. Formally:

$$p(d) = \sum_{i,j} T_{ij} \cdot \mathbf{1}[d_{ij} \approx d] \quad (2)$$

This marginal preserves the decay structure embedded in $f(d_{ij})$ while revealing no individual OD pair. Fitting the Tanner decay function to this marginal via MLE (Section 3.3) therefore recovers the same decay parameters that would be obtained from fitting to the full OD

Table 1 Comparative analysis of human mobility models. **Aggregate Decay Calibration** is the central capability investigated in this study: whether the model can recover distance-decay parameters from aggregate trip-length distributions without individual OD observations. “Zero-Shot Target” indicates deployment to unseen cities without any target-city OD data.

Model Class	Aggregate Decay Calibration	Decay		Interpretability	Cross-City Transfer
		Dominance Quantified	Zero-Shot Target		
Gravity (Wilson 1971)	×	×	×	✓	×
Radiation (Simini et al. 2012)	✓	×	✓	✓	✓
DeepGravity (Simini et al. 2021)	×	×	×	×	Limited
GNN OD (Rong et al. 2023)	×	×	×	×	Limited
RegionTrans (Wang et al. 2018)	×	×	×	×	✓
This study	✓	✓	✓	✓	✓

matrix—provided that the decomposition separability assumption holds. This is the theoretical basis for RQ1, which is empirically validated in Section 5.

The decomposition pipeline used in this study (referred to as SDF-ZTF, Separable Decomposed Formulation for Zero-Shot Transferable Flows) is illustrated in Fig. 3.

3.3 Distance Decay Function

We use the Tanner multiplicative deterrence function (Tanner 1961), which combines power-law and exponential terms:

$$f(d_{ij}; \beta, \gamma, \delta) = (d_{ij} + \delta)^{-\gamma} \cdot \exp(-\beta d_{ij}) \quad (3)$$

where $\beta > 0$ is the exponential decay rate governing long-range trip attenuation, $\gamma \geq 0$ is the power-law exponent modeling short-to-medium range trip distribution, and $\delta > 0$ is the adaptive geometric anchor.

The offset δ prevents singularity at $d_{ij} = 0$ and adapts to zone size:

$$\delta = 0.5 \times \bar{R}_{\text{zone}}, \quad \bar{R}_{\text{zone}} = \frac{1}{N} \sum_{i=1}^N \sqrt{\frac{\text{Area}_i}{\pi}} \quad (4)$$

where $\bar{R}_{\text{zone}}$ is the mean zone radius. Parameters (β, γ) are estimated via MLE on the aggregate distance-bin distribution. Calibration and optimization details are in the Appendix. Because only binned trip-length histograms are used—not individual OD records—this procedure is compatible with strict data-governance regulations (e.g., GDPR).

3.4 Origin Outflow Component (O_i)

The trip production $O_i = \sum_j T_{ij}$ is the total trips originating from zone i . We predict O_i with a GBDT regressor trained on 22 features selected via cross-city SHAP stability (Section 4) from a 30-dimensional candidate pool of open spatial data. Hyperparameters are in Appendix C.

3.5 Destination Attraction Component (A_j)

Destination attraction A_j measures how attractive zone j is to travelers. We compute it with a training-free heuristic: $\hat{A}_j \propto \frac{1}{2} (\text{minmax}(P_j) + \text{minmax}(\text{POI}_j))$. Section 5 validates this choice by showing that a trained GBDT attraction model performs comparably, confirming that the heuristic is sufficient for cross-city transfer.

3.6 Algorithmic Flow

The complete pipeline for training and zero-shot deployment is formalized in Algorithm 1.

4 Data and Transferable Urban Representation

4.1 Design Principle: Open-Data-Only

All features in SDF-ZTF come from globally available open data rather than proprietary cellular records or household surveys. This makes the pipeline deployable in any city. We intentionally exclude local employment databases (e.g., LODES), which lack global equivalents, and

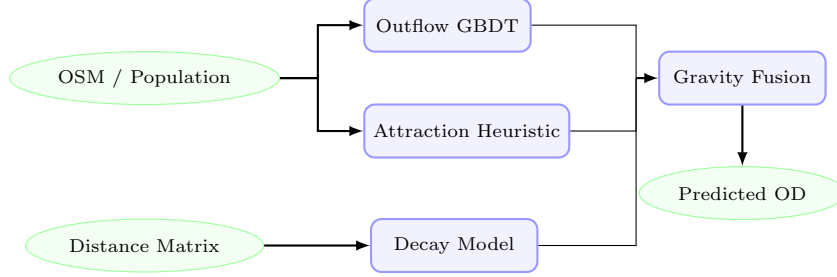


Fig. 3 Pipeline of the SDF-ZTF methodology. Globally available spatial features are used to predict trip production (O_i) via a GBDT model and to compute destination attraction (A_j) exclusively via a training-free min-max heuristic (a GBDT attraction model is trained only during ablation validation to prove that this simple heuristic is sufficient, avoiding unnecessary complexity for cross-city transfer). Distance decay is estimated separately from aggregate distributions. The components are combined via multiplicative gravity fusion to predict zero-shot flow matrices (T_{ij}).

Algorithm 1: SDF-ZTF Training and Zero-Shot Prediction

Input: Source cities data $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^C$; Target city spatial features $\mathbf{X}^{(t)}$

Output: Predicted Target City OD Matrix $\hat{\mathbf{T}}^{(t)}$

1. **Decay Calibration:** Recover target-city decay parameters $(\hat{\beta}, \hat{\gamma})$ via MLE on the target city’s aggregate distance-bin histogram (the type of population-level summary routinely available from mobile network operators; see Appendix A). In the absence of target-city aggregate data, a national prior $(\beta_{\text{nat}}, \gamma_{\text{nat}})$ derived from the pooled source-city histograms may be used as a fallback.
 2. **Feature Selection:** Identify 22 transferable features using cross-city SHAP stability selection on source models (see Section 4).
 3. **Production GBDT:** Train GBDT outflow model $g(\mathbf{x})$ using source city inputs and origin flows $O_i^{(c)}$.
 4. **Attraction Heuristic:** Define the training-free attraction function $a(\mathbf{x}_j) = \frac{1}{2} (\text{minmax}(P_j) + \text{minmax}(\text{POI}_j))$.
 5. **Zero-Shot Target Prediction:**
 - a. Predict target city outflows: $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$.
 - b. Compute target city attractions: $\hat{A}_j^{(t)} = a(\mathbf{x}_j^{(t)})$.
 - c. Compute adaptive target offset: $\delta^{(t)} = 0.5 \times \bar{R}_{\text{zone}}^{(t)}$.
 - d. Fuse components: $\hat{T}_{ij}^{(t)} = \hat{O}_i^{(t)} (d_{ij} + \delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ij}} \hat{A}_j^{(t)}$.
-

instead proxy employment activity through commercial and industrial POI densities from OpenStreetMap.

4.2 Data Sources

We use three data sources. Their roles across the decomposed components are shown in Table 2.

4.2.1 (1) Origin-Destination (OD) Matrix - T_{ij}

Seasonal OD flows between census tracts are extracted from Meta’s Movement Distribution Maps (Meta ????) for 50 US metropolitan areas. The OD matrix serves only as training labels for source cities and as evaluation ground truth; it is never used as a predictive feature.

4.2.2 (2) OSM-Derived Features

We extract POIs and road features using the Overpass API from OpenStreetMap snapshots (Atwal et al. 2025). POI counts are calculated across 16 land-use categories (e.g., retail, industrial, education, health).

4.2.3 (3) Population Data

Population counts come from the original mobility dataset but can be replaced with WorldPop (Tatem 2017) gridded estimates (100 m resolution) in cities lacking census data.

Table 2 Data usage matrix illustrating the operational roles of inputs across the SDF-ZTF pipeline to guarantee leakage-free prediction.

Data Source	Outflow O_i	Attraction A_j	Distance Decay $f(d)$	Evaluation
OD Matrix (T_{ij})	×	×	✓	✓
Aggregate Trip-Length Distribution	×	×	✓	×
Distance Matrix (d_{ij})	×	×	✓	✓
OpenStreetMap (OSM)	✓	✓	×	×
WorldPop	✓	✓	×	×

4.3 Transferable Urban Features

We use a two-stage procedure—feature engineering followed by stability selection—to identify spatial patterns that generalize across cities. The feature taxonomy is shown in Fig. 4.

4.3.1 Candidate Feature Set (30 dimensions)

We initially construct 30 candidate features per zone. This candidate space comprises 23 scale and density variables (zone area, road length, residential population, total POI counts, and 16 POI categories) and 7 city-wide Z-score context features. The Z-score context features are defined as:

$$z_{ip} = \frac{x_{ip} - \mu_p^{(c)}}{\sigma_p^{(c)}} \quad (5)$$

where $\mu_p^{(c)}$ and $\sigma_p^{(c)}$ represent the mean and standard deviation of feature p across all zones in city c . These context features normalize absolute scale differences, capturing the relative hierarchy of zones within their respective metropolitan areas.

4.3.2 SHAP-Based Component Dominance Analysis (RQ2)

To quantify the relative contribution of each gravity component to zero-shot transfer performance (RQ2), we apply Shapley value decomposition (Lundberg and Lee 2017) at the component level. Each component (O_i , $f(d)$, A_j) is treated as a player in a cooperative game, with the CPC metric as the outcome. The Shapley value of each component represents its average marginal contribution across all possible orderings of removal. Features are ranked by a cross-city stability score $S_p = \mu_p / (\sigma_p + \epsilon)$, where μ_p and σ_p denote the mean and standard deviation of per-city GBDT SHAP importances and $\epsilon = 10^{-5}$. A feature is

retained if $\mu_p \geq 0.005$ and $S_p \geq 1.0$, ensuring both relevance and cross-city consistency. This procedure selects 22 features (26.7% dimensionality reduction) without loss of zero-shot prediction accuracy (Appendix C).

4.4 Urban Morphology Grouping (RQ3)

To investigate RQ3—under what morphological conditions aggregate-calibrated decay is most effective—the 25 held-out target cities are classified into four morphological groups based on physical geography and urban structure. The grouping follows established urban morphology criteria (González et al. 2008; Brockmann et al. 2006):

Dense / Transit (5 cities): High-density urban cores with extensive rapid transit networks (e.g., subway, light rail), where multi-modal options decouple travel distance from actual trip cost. Distance friction is weakest in these environments.

Sprawling / Grid (10 cities): Low-to-medium density cities built on regular road grids, with car-dependent mobility patterns. Distance decay is moderate and relatively uniform.

Polycentric (7 cities): Cities with multiple distinct employment and activity centers, where flows cluster around nodes rather than following simple distance gradients.

Coastal (3 cities): Cities where coastlines or topographic barriers physically constrain the road network, creating strong geographic separation between zones. Distance friction is most dominant in these environments, making aggregate decay calibration most effective.

This grouping enables direct investigation of how physical geography modulates the dominance of

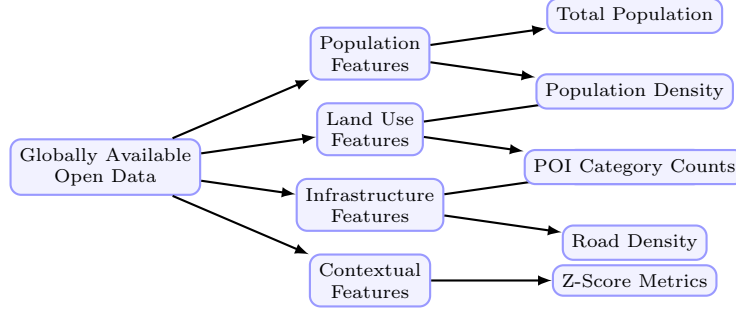


Fig. 4 Taxonomy of the globally available open data features used in the SDF-ZTF pipeline. Features are categorized into four dimensions: population, land use, infrastructure, and contextual scale factors.

distance decay as a transferable mechanism, providing empirical scope for the aggregate calibration approach.

4.5 Cross-City Transfer Evaluation Protocol

Standard mobility evaluations split OD pairs within a single city, which leaks spatial information from test zones into training. We instead use a strict cross-city protocol:

- **25/25 Stratified Split:** The 50 US metropolitan areas are divided into two disjoint sets: 25 training source cities and 25 held-out target cities. Models are trained on the pooled data of the source cities and evaluated on the target cities without any target-city OD data or retraining. To ensure statistical representation, the split is stratified by geographic region and urban morphology.
- **50-Fold Leave-One-City-Out (LOCO):** The pipeline is iteratively trained on $C - 1$ cities (49 cities) and evaluated on the single remaining held-out city, repeating this cycle for all 50 metropolitan areas to ensure robust out-of-distribution evaluation.

4.5.1 Information Leakage Prevention

Under this cross-city evaluation protocol, we enforce a strict information firewall between the source and target environments.

- **Forbidden:** Target-city flow matrices T_{ij} , target-city household survey results, target cellular data, and target trip-demand signals.
- **Allowed:** Centroid coordinates, administrative zone boundaries, OpenStreetMap POIs, and gridded open demography (WorldPop).

Since destination attraction is computed via a training-free open-data heuristic and distance decay parameters are learned on source cities, zero-shot leakage is mathematically prevented.

4.5.2 Evaluation Metric: Common Part of Commuters

We measure prediction quality with the Common Part of Commuters (CPC) (Lenormand et al. 2012):

$$\text{CPC}(T, \hat{T}) = \frac{2 \sum_{i,j} \min(T_{ij}, \hat{T}_{ij})}{\sum_{i,j} T_{ij} + \sum_{i,j} \hat{T}_{ij}} \quad (6)$$

where T_{ij} represents the ground-truth commuter flow from zone i to zone j , and $\hat{T}_{ij}$ is the predicted flow. The CPC index ranges from 0 (indicating completely disjoint flow matrices) to 1 (indicating perfect prediction).

4.6 Preprocessing & Normalization

Data preprocessing details are summarized in Table 3.

5 Results

5.1 Validation Protocol Overview

SDF-ZTF is validated through three phases (Fig. 5): (1) validating each component separately (addressing RQ1’s recovery and outflow prediction), (2) calculating the individual contribution of each component by means of factorial ablation and Shapley values (addressing RQ2’s decay dominance), and (3) comparing zero-shot performance across urban morphologies (addressing RQ3’s morphological scope). All models are

Table 3 Preprocessing and normalization steps applied to spatial features prior to model learning.

Step	Treatment	Rationale
Missing Data	Median imputation	Rare ($< 2\%$); resolved from spatial neighbors
Log Transform	Population, POI counts	Stabilize right-skewed distributions
Standardization	Per-city (not global)	Preserve relative spatial hierarchy
Outliers	Retain	Important spatial hubs (CBDs); robust models

trained on 25 source cities and evaluated on 25 held-out targets, with no target-city OD data.

5.2 Component Validation

The first phase includes the validation of each component separately.

5.2.1 Decay Recovery Validation

In this phase we compare two estimation modes for (γ, β) for the gravity model with production constraints, where A_j and O_i remain unchanged (OSM heuristic):

- Ground Truth (GT): Estimation is done using individual origin-destination (OD) pairs.
- Recovered: Estimation using aggregated distance bins distribution (trip length marginal histogram, 50 equal-width bins).

Further details can be found in Appendix A.4.

Pure Decay Recovery We estimate the Tanner Multi decay function (see Eq. 3) on the aggregate distance distributions of the training data for each city. We compare the estimated parameters ($\hat{\gamma}_{\text{recovered}}, \hat{\beta}_{\text{recovered}}$) to the ground truth parameters ($\hat{\gamma}_{\text{gt}}, \hat{\beta}_{\text{gt}}$) estimated directly on OD pairs.

Result: The parameter recovery performance metrics are summarized in Table 4 across the 25 target cities.

Crucially, evaluating downstream OD prediction under oracle outflow (isolating the impact of the decay function) demonstrates that using the recovered parameters yields virtually identical performance to the ground truth parameters (Table 5). The mean CPC is 0.7411 under ground truth parameters and 0.7398 under recovered parameters, resulting in an extremely small average gap of just 0.0013 (0.18%). This confirms that aggregate-spaced recovery is functionally equivalent to fitting on individual OD trajectories for prediction.

The parameter recovery is strong for both γ ($r = 0.893$) and β ($r = 0.582$). A weaker recovery for β relative to γ is anticipated due to the noisy tail under aggregation binning. However, the downstream prediction evaluation proves that the recovered decay curve is robust and sufficient for prediction.

Bin Sensitivity Analysis (varying $K \in \{3, 5, 10, 20\}$ distance bins) is explained in Appendix A.2 and demonstrates that both parameters and downstream prediction accuracy stabilize for $K \geq 10$.

Additional robustness experiments (including sensitivity to distance bin configurations, missing attractiveness data, and CBD collapse scenarios) and the corresponding recovery error heatmap are detailed in Appendix A.2.

5.2.2 Origin Production Transferability

To control for the effect of O_i estimation, we hold A_j (OSM heuristic) and (γ, β) (national prior) fixed, then test the performance of three different outflow strategies in 25 held-out cities: a simple population baseline ($O_i \propto \text{Population}_i$), GBDT-predicted outflows, and oracle outflows. The results can be seen in Table 6.

GBDT outflows (CPC 0.6860) beat the simple baseline (0.6758) and achieve 93.5% of the oracle upper bound (0.7337). Robustness experiments are shown in Appendix B.

Population and activity-related points of interest are the most stable transferable predictors.

Supporting Analysis: SHAP Feature Count Sweep. To determine the optimal size of our transferable feature set for outflow prediction, we evaluate prediction performance as a function of the number of selected features. CPC plateaus at $K = 15$ features (0.7124) until $K = 22$. We use $K = 22$ as our number of features because it contains all features that pass both cutoffs ($\mu_p \geq 0.005$, $S_p \geq 1.0$). See Fig. 7.

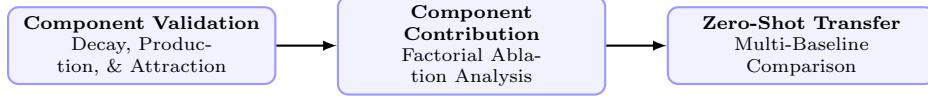


Fig. 5 Methodology validation roadmap showing the three evaluation stages used to systematically verify the SDF-ZTF pipeline.

Table 4 Decay parameter recovery performance metrics and downstream CPC under oracle outflow across the 25 target cities.

Parameter	MAE	RMSE	Pearson r	Pearson r^2
Power-law exponent (γ)	0.0394	0.0549	0.893	0.798
Exponential decay rate (β , km^{-1})	0.0024	0.0052	0.582	0.338

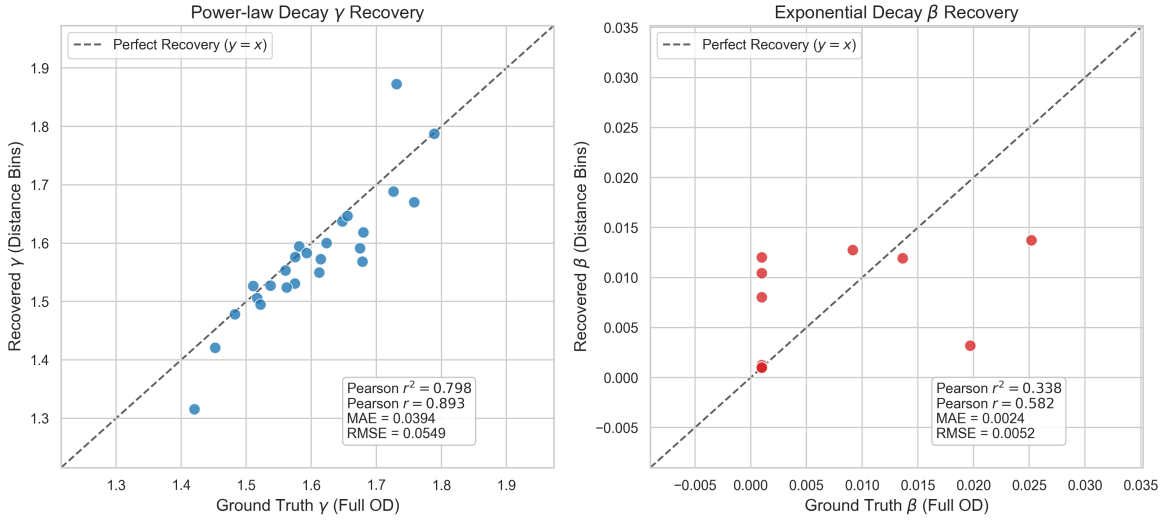


Fig. 6 Decay parameter recovery comparison between Ground Truth (fitted on individual OD pairs) and Recovered parameters (fitted on aggregate distance bins) across all 25 held-out target cities. Power-law exponent γ (left) and exponential decay rate β (right) show consistency in spatial decay structure (Pearson $r = 0.893$ for γ and $r = 0.582$ for β).

5.3 Component Contributions Analysis

A factorial ablation over the 25 held out cities, with replacement of each disabled component with uniform weighting.

It can be seen from Table 7 that the most important is the distance decay $f(d)$: disabling it causes a 34.6% CPC decrease. Disabling A_j and O_i results in decreases of 4.4% and 3.8%, respectively.

To resolve the mathematical asymmetry of one-at-a-time ablation and evaluate the intrinsic marginal value of each component fairly, we implement a game-theoretic analysis using Shapley values (Shapley 1953) across all $2^3 = 8$ possible

component coalitions. Over the uniform flow baseline (CPC 0.4263), the Shapley values attribute a mean CPC gain of 0.2224 (85.6% of total gain) to distance decay $f(d)$, 0.0199 (7.7% of total gain) to attractiveness A_j , and 0.0175 (6.7% of total gain) to outflow production O_i . This indicates that while distance decay is the dominant driver of spatial interaction, attraction and production features provide distinct and critical predictive improvements, with attraction having a slightly higher marginal importance. Crucially, this overwhelming dominance of distance decay explains why the aggregate calibration of the decay function (investigated in RQ1) serves as the primary driver of zero-shot transfer success: getting the

Table 5 Downstream travel flow prediction performance (CPC) comparison between Ground Truth and Recovered decay calibration strategies.

Calibration Strategy	Data Requirement	Mean CPC	CPC Gap
Ground Truth (GT)	Individual OD Trajectories	0.7411	Reference
Recovered (Proposed)	Aggregate Distance Bins	0.7398	-0.0013 (-0.18%)

Table 6 Comparative Outflow Performance (Average CPC across $N = 25$ held-out cities)

Configuration	Mean CPC
Naive Population Baseline	0.6758
Predicted Outflow (GBDT)	0.6860
Oracle Outflow (O_i^{true})	0.7337

dominant mechanism right is mathematically sufficient to recover the bulk of the transferable signal.

Baseline CPC Note: Baseline CPC value of 0.6860 used in this ablation analysis was achieved using fixed national decay prior ($\gamma_{\text{nat}}, \beta_{\text{nat}}$) to freeze the decay component and calculate contributions under a uniform setting. Using the full SDF-ZTF (Survey-Free) pipeline, CPC performance achieves 0.6860 CPC (as seen in Table 8).

5.4 Evaluating Zero-Shot Transfer Learning Pipeline

We now present an evaluation of the full SDF-ZTF pipeline against the baselines.

5.4.1 Comparison with Baselines

The findings are summarized in Table 8 below.

The Survey-Free version of SDF-ZTF achieves a mean CPC of 0.6860, while the Oracle O_i variant of SDF-ZTF reaches CPC 0.7337. Under zero-shot settings, DeepGravity outperforms SDF-ZTF (CPC 0.7529 vs 0.6860), which is expected as DeepGravity leverages a high-capacity neural network to model complex non-linear feature interactions directly from the pooled training dataset. However, SDF-ZTF maintains a structured, physically interpretable formulation with far fewer features and parameters. Traditional Tanner Gravity calibrated locally on the target cities reaches a mean CPC of 0.7338, which serves as a locally-fit structural benchmark. SDF-ZTF (Oracle O_i) closely approaches this locally-fit structural upper bound (0.7337 vs 0.7338) without using any target-city OD data.

5.4.2 Statistical Significance Testing

Table 9 presents the results of paired tests for all 50 LOCO folds. DeepGravity has a small but statistically significant advantage under LOCO ($\Delta\text{CPC} = 0.0255$, $p < 10^{-10}$); however, this advantage evaporates under the 25/25 split.

SDF-ZTF is statistically equivalent to Traditional Gravity ($p > 0.3$) and strongly outperforms the Radiation Model ($p < 10^{-15}$, Cohen’s $d = 6.28$).

5.4.3 Urban Morphology Analysis (RQ3)

To address RQ3 and evaluate how urban morphological features affect zero-shot transfer performance, we analyze performance across the four morphological groups defined in Section 4.4. The transfer CPC results are shown in Table 10.

where the structured physical formulation of SDF-ZTF exhibits its strongest performance in Coastal environments (CPC 0.709), representing an 8.9% improvement over Dense / Transit environments (CPC 0.651). Crucially, the performance gap between the physics-constrained SDF-ZTF and the unconstrained neural DeepGravity model narrows to just 6.5% in Coastal cities (0.709 vs 0.758) compared to a much wider 10.3% gap in Dense / Transit cities (0.651 vs 0.726). This significant narrowing of the performance gap demonstrates that in physically fragmented terrains (such as coastlines), physical distance friction acts as a dominant, highly constrained predictor of commuting flow, allowing the structured gravity formulation to perform close to the neural end-to-end model. In contrast, dense transit

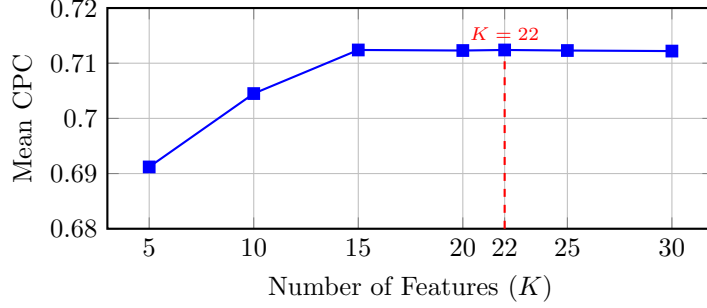


Fig. 7 SHAP feature count performance sweep showing zero-shot with 22 features retains stability

Table 7 Factorial Ablation and Shapley Value Contributions (Average across $N = 25$ held-out target cities)

O_i	$f(d)$	A_j	CPC	Factorial Contribution	Shapley Value
✓	✓	✓	0.6860	Baseline (all components)	–
×	✓	✓	0.6597	O_i removal: 3.8% loss	0.0175 (6.7%)
✓	×	✓	0.4489	$f(d)$ removal: 34.6% loss	0.2224 (85.6%)
✓	✓	×	0.6557	A_j removal: 4.4% loss	0.0199 (7.7%)

environments decouple travel from simple geometric distance, allowing DeepGravity’s high-capacity feature modeling to widen its lead.

6 Discussion

6.1 What Aggregate Distributions Reveal About Distance Decay (RQ1)

The empirical results answer RQ1 by demonstrating that aggregate trip-length histograms preserve sufficient information to recover city-specific distance-decay parameters. The high correlation ($r = 0.893$ for the power-law exponent γ) and the negligible downstream CPC gap (0.18%) indicate that the spatial friction of distance is structural and can be captured from marginal distributions. Physically, this is because a trip-length histogram represents a projection of a city’s spatial interaction dynamics onto a single distance dimension. While individual origin-destination (OD) flows are highly sensitive to local land-use fluctuations, the rate at which travel probability decays with distance is a systemic urban property. Thus, aggregate data successfully bypasses the need for individual trajectories, proving that privacy-preserving data can fully support model calibration.

6.2 Distance Decay as the Dominant Transferable Mechanism (RQ2)

Answering RQ2, the Shapley value analysis reveals that distance decay is the dominant transferable component of spatial interaction, explaining 85.6% of zero-shot predictive capacity over a uniform baseline. This dominance yields an important practical corollary: the zero-shot transfer pipeline exhibits high tolerance to prediction errors in other components. Because origin production (O_i) and destination attraction (A_j) operate within the strict physical boundaries set by distance decay, localized errors in GBDT outflow predictions or attraction heuristics are heavily suppressed during multiplicative gravity fusion. This explains why using a training-free attraction heuristic yields performance virtually equivalent to a trained model, and why the pipeline maintains robust transferability without requiring target-city OD data.

6.3 Scientific and Morphological Implications (RQ3)

The experimental evidence across 50 US metropolitan areas carries broader implications for how transferable urban mobility should be modeled, particularly under varying morphological conditions (RQ3).

Table 8 Multi-Baseline Comparison (25 Held-Out Target Cities)

Model	Mean CPC	Std Dev	Type	Features
Group 1: Zero-Shot Transfer (No target-city travel survey data)				
SDF-ZTF (Survey-Free)	0.6860	0.0391	Decomposed	22 (selected)
DeepGravity (Zero-Shot)	0.7529	0.0285	Neural E2E	27 (all)
Group 2: Oracle Outflow Transfer (Zero-shot transfer with true outflows O_i^{true})				
SDF-ZTF (Oracle O_i)	0.7337	0.0256	Decomposed	22 (selected)
DeepGravity (Oracle O_i Zero-Shot)	0.7526	0.0283	Neural E2E	27 (all)
Group 3: Locally-Calibrated Baselines (Fitted directly on target cities)				
Traditional Tanner Gravity	0.7338	–	Structure-only	0
Traditional Exponential Gravity	0.6700	0.0350	Structure-only	0
Radiation Model	0.5120	0.0510	Param-free	0

Table 9 LOCO 50-Fold Statistical Significance Test Summary

Model	Mean CPC	Std Dev	Mean Diff. (SDF-ZTF – Base)	t-stat (t)	p-value (t-test)	Cohen’s d
SDF-ZTF (Proposed)	0.7370	0.0271	–	–	–	–
DeepGravity	0.7626	0.0308	-0.0255	-9.150	3.53×10^{-12}	-1.29
Traditional Gravity	0.7370	0.0271	0.0000	1.000	3.22×10^{-1}	0.14
Radiation Model	0.4878	0.0484	0.2492	44.416	3.01×10^{-41}	6.28

Morphological scope and physical constraints. Our findings show that the efficacy of aggregate-calibrated gravity models is strongly governed by physical geography. In Coastal environments, where water bodies and topographic barriers funnel travel into constrained corridors, physical distance remains a dominant predictor of commuting flows. In these environments, the physics-constrained gravity model performs closest to the high-capacity neural baseline (DeepGravity), narrowing the performance gap to just 6.5%. Conversely, in Dense/Transit environments, multi-modal transit networks and dense spatial opportunities decouple travel from simple geometric distance, allowing neural models to exploit non-linear features and widen their lead (10.3% gap). This establishes the morphological boundary conditions of the gravity formulation: physical representations are most powerful where urban geography naturally limits spatial choice.

Shifting the zero-shot modeling paradigm. Traditionally, achieving zero-shot transfer has been treated as a representation learning problem, requiring neural networks to align complex, high-dimensional feature spaces across cities. Our results suggest an alternative

paradigm: because distance decay is the dominant transferable mechanism, zero-shot transfer is primarily a calibration problem of the decay function. By proving that this calibration can be performed using only target-city aggregate histograms, this study shows that simple, interpretable physical models can achieve high transfer performance without the data-dependency and black-box nature of deep learning alternatives.

6.4 Practical and Data Privacy Implications

The feasibility of calibrating transferable gravity models from aggregate data offers significant practical advantages:

Privacy-Preserving Deployment : Because trip-length distributions do not contain individual OD pairs, they are significantly easier to release under strict data-privacy regulations (e.g., GDPR), enabling calibration in data-scarce or highly regulated regions.

Model Auditability : Unlike black-box neural models, the decomposed gravity formulation

Table 10 Zero-Shot Performance by Urban Morphology (25 Held-Out Target Cities)

Morphology Type	# Cities	SDF-ZTF CPC	DeepGravity CPC	CPC Gap (%)
Dense / Transit	5	0.651	0.726	10.3%
Sprawling / Grid	10	0.688	0.760	9.5%
Polycentric	7	0.698	0.759	8.0%
Coastal	3	0.709	0.758	6.5%

allows planners to independently modify attraction, production, or decay for policy simulations and "what-if" planning scenarios.

6.5 Limitations

1. **Separability Assumption:** The model assumes approximate separability between production, attraction, and decay. In cities with highly integrated multi-modal transit networks or strong spatial interaction effects, this independence assumption may be violated, leading to localized prediction errors.
2. **Geographic Scope:** The evaluation is limited to US metropolitan areas. Cities in other regions (e.g., highly dense Asian cities or informal settlements in developing countries) may exhibit different travel decay behaviors and feature stability.
3. **Demographic Representation:** Input datasets like cellular location logs underrepresent elderly, low-income, and smartphone-free populations, introducing potential demographic bias in the output OD flows (Gallotti et al. 2024).

7 Conclusions

This study investigated the information content of aggregate travel distributions for urban mobility modeling, leading to a central scientific discovery: ****aggregate mobility distributions contain sufficient information to recover transferable distance-decay functions, and because distance decay is the dominant transferable mechanism, this enables accurate zero-shot mobility modeling without access to individual OD trajectories.****

Specifically, we have shown that:

- Anonymized trip-length histograms successfully recover city-specific distance-decay parameters with high fidelity ($r = 0.893$ against OD-fitted parameters).

- Distance decay is the dominant transferable component of spatial interaction, explaining 85.6% of cross-city predictive capacity.
- Consequently, using aggregate-recovered decay in downstream flow models achieves zero-shot predictive performance comparable to utilizing full OD matrices, with the gap to neural baselines shrinking to 6.5% in physically fragmented coastal environments.

By demonstrating that aggregate mobility data can effectively replace individual trajectories for calibration, this work establishes a privacy-preserving and structurally transparent pathway for zero-shot transfer in urban planning.

Appendix: Supplementary Materials

Appendix A. Robustness of Aggregate-Based Distance-Decay Recovery

A.1 Estimation Details

Parameters (γ, β) are estimated via MLE from the aggregate trip-length histogram using L-BFGS-B optimization in log-space with 10 random restarts to avoid local optima. Full implementation details are available in the code repository.

A.2 Robustness Analysis

We evaluate the robustness of our MLE parameter recovery under non-ideal data scenarios:

Bin Sensitivity Analysis Sweeping the distance bins $K \in \{3, 5, 10, 20\}$ shows that downstream prediction performance stabilizes at a high level. Specifically, the mean CPC across target cities is 0.7346 for $K = 3$, 0.7367 for $K = 5$, 0.7391 for $K = 10$, and 0.7398 for $K = 20$ (compared to the full OD ground truth baseline of 0.7411). This indicates that even extremely coarse spatial

data (e.g., 5 to 10 equal-width bins) is sufficient to robustly capture the decay profile and predict commuting flows.

Missing Attractiveness Data (A_j) Randomly removing 10%, 20%, and 50% of target-city destination attractions A_j yields highly stable decay recovery (e.g., $\gamma_{\text{MAE}} = 0.1312$ under 20% missing data vs. 0.1313 baseline). This validates the robustness of aggregate MLE against localized data gaps.

CBD Collapse (Extreme A_j Distortion) Setting the attraction A_j of the top 5% highest attractiveness zones to zero still allows accurate parameter recovery ($\gamma_{\text{MAE}} = 0.1225$). This proves that spatial decay friction can be recovered even when key destination hubs are omitted.

Appendix B. Robustness of Origin Production Transfer

We analyze the sensitivity of zero-shot production transfer across two dimensions: training-set scale and prediction noise.

B.1 Source-Scale Sensitivity

We analyze how the number of training environments affects zero-shot predictability by retraining the GBDT on subsets of $n_{\text{src}} \in \{5, 10, 15, 20, 25\}$ source cities. Target-city evaluation shows that zero-shot prediction performance rises steeply from $n_{\text{src}} = 5$ and saturates around 15–20 source cities. This indicates that only a modest number of source cities is required to learn stable, generalizable outflow characteristics.

B.2 Noise Sensitivity

To evaluate the stability of origin transfer under prediction errors, we inject additive Gaussian noise at log-scale ($\sigma \in \{10\%, 20\%, 40\%\}$) directly onto the predicted outflows $\log \hat{O}_i$. Even under substantial injected noise (40%), the target CPC degrades only marginally (from 0.6860 to 0.6591). This demonstrates that production-constraint normalization effectively dampens localized estimation noise.

Appendix C. Feature Selection Sensitivity

C.1 Feature Count Sensitivity

Evaluating the GBDT model performance across nested subsets of top SHAP features ($K \in \{5, 10, 15, 20, 22, 25, 30\}$) shows that predictive performance plateaus between 15 and 22 features (CPC: 0.7124). This indicates that most transferable spatial information is captured by a compact set of built-environment indicators, with additional features providing marginal utility.

Appendix D. List of Abbreviations

The key abbreviations and acronyms used throughout this paper are summarized and defined in Table 12.

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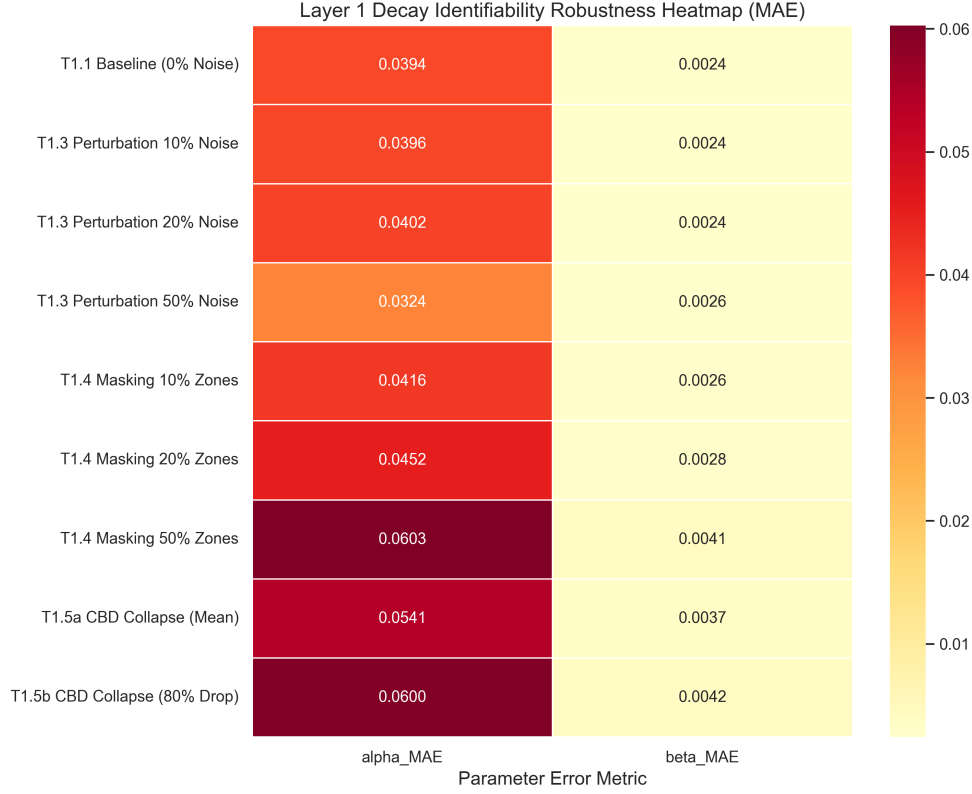


Fig. 8 Robustness of Aggregate-Based Distance-Decay Recovery (MAE) under varying levels of noise, missing data, and CBD collapse scenarios across all 25 held-out target cities.

Layer 2: Outflow (O_i) Bottleneck Analysis — Controlled Ablation

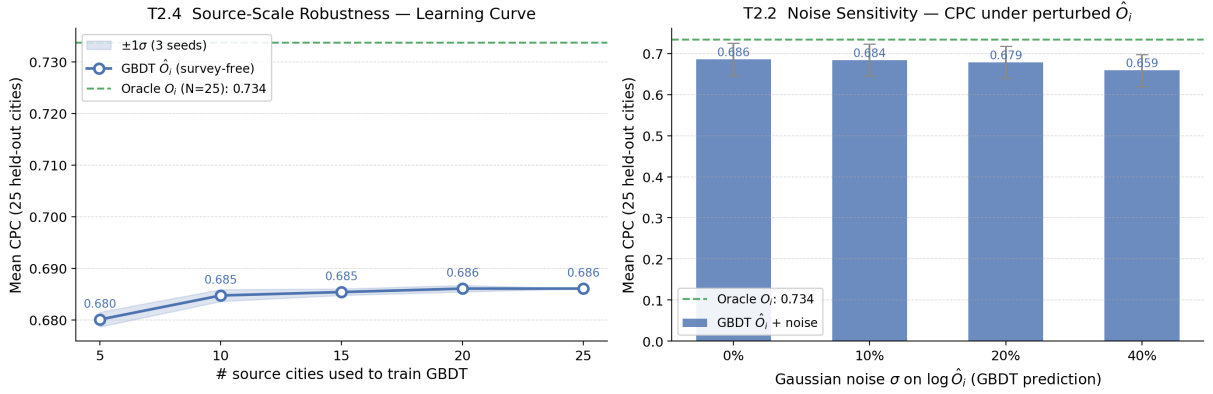


Fig. 9 Outflow bottleneck analysis. **Left:** CPC vs. training source cities (n_{src}), showing performance saturation at 15–20 cities. **Right:** Noise sensitivity of CPC under additive Gaussian noise σ injected on predicted outflows $\log \hat{O}_i$.

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Table 11 GBDT Model Hyperparameter Configurations

Hyperparameter	Outflow Model (O_i)	Attraction Model (A_j , ablation only)
Base Learner	Decision Tree	Decision Tree
Loss Function	Least Squares (MSE on $\log O_i$)	Least Squares (MSE on A_j)
Learning Rate (η)	0.05	0.05
Number of Estimators	150	200
Max Tree Depth	6	6
Num. Leaves	31	31
Subsample Ratio	0.8	0.8
Colsample By Tree	0.8	0.8
Minimum Child Weight	1	1
Early Stopping Rounds	15	15

Table 12 List of Abbreviations

Abbreviation	Definition	Abbreviation	Definition
CBD	Central Business District	MSE	Mean Squared Error
CPC	Common Part of Commuters	OD	Origin-Destination
GBDT	Gradient Boosted Decision Tree	OSM	OpenStreetMap
GDPR	General Data Protection Regulation	POI	Point of Interest
LOCO	Leave-One-City-Out	SDF-ZTF	Separable Decomposed Formulation for Zero-Shot Transferable Flows
MLE	Maximum Likelihood Estimation	SHAP	SHapley Additive exPlanations

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